

# Complete Small-Argument Asymptotics for the Inverse Fabius Function

Closed Lambert–Bell coefficients, Gaussian partition sums,  
and an exact 2-adic Mellin bridge

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28 August 2026

Repository snapshot ad82c27ffc6f90b3406f46c130c86d3cb83c6225

## Abstract

Let  $F$  be the Fabius distribution function on  $[0, 1]$  and  $G = F^{-1}$  its increasing inverse. The repository corpus already contains a sharp all-orders small- $x$  expansion of  $\log F(x)$  in the exact lower-Lambert coordinate, including a nonconstant one-periodic Gamma–zeta fluctuation, and it contains a triangular phase-reversion algorithm for  $G(y)$  as  $y \downarrow 0$ . The purpose of this report is therefore not to repeat that machinery, but to close several remaining coefficient-level gaps.

Write

$$T = \log(1/y), \quad L = \log 2, \quad \rho = \sqrt{2T/L}, \quad \vartheta = \rho + \frac{1}{L} + \frac{1}{2}.$$

The leading inverse term is

$$G_0(y) = \frac{\sqrt{T}}{e\sqrt{L}} \exp(-\sqrt{2LT}).$$

We prove that, for every  $N$ ,

$$\frac{G(y)}{G_0(y)} = \sum_{n=0}^{N-1} \frac{g_n(\log \rho, \vartheta)}{\rho^n} + \mathcal{O}_N\left(\frac{(1 + \log \rho)^N}{\rho^N}\right),$$

uniformly in the fast phase. Every  $g_n$  is a polynomial in  $\log \rho$  whose coefficients are smooth one-periodic differential polynomials in the forward Gamma–zeta phase and the forward saddle coefficients. Beyond the existing recursive construction, we give a nonrecursive generalized Lagrange–Bürmann coefficient formula for the phase coefficients  $d_n$ , and a direct partial-Bell-polynomial formula for  $g_n$ . These formulas make the finite dependence at order  $n$  explicit and yield, in particular,

$$g_n(\log \rho, \vartheta) = \frac{(\log \rho)^n}{2^n n!} + \mathcal{O}((1 + \log \rho)^{n-1}),$$

with a universal highest logarithm independent of the periodic fluctuation. The third inverse correction is displayed explicitly. An all-orders formula for the endpoint elasticity  $yG'(y)/G(y)$  follows by a two-scale differential operator.

At the forward saddle level we derive a second closed formula: the complete Gaussian correction at order  $n$  is a finite constrained partition sum over multiplicities  $m_0, m_3, m_4, \dots$ . It includes the Bromwich denominator and specializes to the classical Mills-ratio coefficients when all higher saddle cumulants vanish.

Finally, for the logarithm  $K(t)$  of the defining Laplace transform, we prove the exact identity

$$K(t) = -\frac{\log^2 t}{2L} + \frac{1}{2} \log t - C_B + \Psi(\log_2 t) + B(t),$$

where

$$B(t) = -\sum_{k \geq 0} \log(1 - e^{-2^k t}) = \sum_{n \geq 1} \frac{2^{\nu_2(n)+1} - 1}{n} e^{-nt}.$$

Thus one and the same Mellin kernel  $\Gamma(s)\zeta(s+1)/(1-2^{-s})$  produces the quadratic logarithm, the Gamma-zeta periodic phase, the Bernoulli power corrections, and a large- $t$  family of explicitly 2-adic exponentially small sectors. This supplies a concrete starting point for an exponentially improved inverse theory. All claims marked as new are relative to the inspected repository snapshot; no claim of global publication priority is made.

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# 1 Executive summary and nonduplication boundary

## 1.1 Corpus audit

The recursive audit was performed at commit `ad82c27ffc6f90b3406f46c130c86d3cb83c6225` of `VladimirReshetnikov/ProveIt`. The current documentation tree contains living expositions, consolidated frontier volumes with provenance ledgers, frozen historical variants, a mathematical glossary, a Lean walkthrough, formalization audits, transcribed primary papers, and many subject-specific reports. Repeated drafts were read as mathematical lineages rather than as independent sources; where a consolidated volume superseded its absorbed drafts, the consolidated statement was used.

The most important nonduplication boundary is the current file

```
non-formalized-research-frontiers/drafts/inverse-and-sampling/
Inverse_and_Sampling_Frontiers/Inverse_and_Sampling_Frontiers.tex
```

It already establishes the first two phase-aware inverse corrections, proves that no phase-independent second term can exist, gives the two-term endpoint elasticity, and supplies a triangular all-orders Lambert–Bell recursion. The formal completion audit additionally records the machine-checked all-orders forward expansion in the exact Lambert phase. Those results are imported here and are not presented as new.

The formula-level searches used to set the novelty boundary included the exact inverse phase, Bell-polynomial reversion, Gaussian saddle contractions, Mellin kernels, 2-adic valuations, Touchard polynomials, and candidate Dirichlet series. In particular, the current corpus contains the spectral Dirichlet series  $\zeta(s)/(1 - 2^{-s})$  for sinc-zero multiplicities, but the coefficient sequence  $(2^{\nu_2(n)+1} - 1)/n$  and the exact large- $t$  expansion of  $B(t)$  below were not found.

## 1.2 What is added here

| Topic                               | Result                                                                                                                                                  | Status        |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Endpoint inverse coefficients       | A bivariate Lagrange–Bürmann extractor for every phase coefficient $d_n$ , replacing triangular iteration by a finite nonrecursive coefficient formula. | New           |
| Multiplicative inverse coefficients | A direct partial-Bell formula for every $g_n$ in $G/G_0$ , not merely the exponential-Bell conversion from logarithmic coefficients.                    | New           |
| Universal logarithms                | $g_n = (\log \rho)^n / (2^n n!) +$ lower logarithmic degree, uniformly in phase.                                                                        | New           |
| Third correction                    | Explicit $d_3, h_3, g_3$ in terms of $\Psi, A_1, A_2$ and their derivatives.                                                                            | New           |
| Inverse derivatives                 | An arbitrary-order coefficient formula for $yG'(y)/G(y)$ and an operator calculus for repeated logarithmic derivatives.                                 | New           |
| Forward saddle coefficients         | A constrained-partition sum for the complete CDF saddle correction $C_n$ , including the $1/z$ Bromwich factor.                                         | New formula   |
| Dyadic completion                   | Exact 2-adic exponential coefficients, Mahler equation, Dirichlet series, Mellin transform, Touchard jets, and exact saddle-map correction.             | New synthesis |
| First two oscillatory inverse terms | The phase shift, $h_1, h_2$ , limit set, and two-term elasticity.                                                                                       | Imported      |
| All-orders forward expansion        | Exact Lambert coordinate, periodic $\Psi$ , and periodic forward coefficients $A_j$ .                                                                   | Imported      |

The central answer to the user's coefficient question is [theorems 3.1, 4.1 and 4.4](#). Together they give a finite, implementable formula for the coefficient at any prescribed order. The remaining sections explain how the forward inputs themselves can be constructed and how the same Mellin object exposes exponentially small dyadic sectors.

## 2 The imported sharp forward expansion

### 2.1 Fabius as a probability distribution

Let  $U_1, U_2, \dots$  be independent uniform random variables on  $[0, 1]$  and put

$$X := \sum_{j \geq 1} 2^{-j} U_j. \quad (2.1)$$

Then  $X \in [0, 1]$  almost surely and the standard Fabius function is

$$F(x) = \mathbb{P}(X \leq x), \quad 0 \leq x \leq 1. \quad (2.2)$$

It is strictly increasing on  $(0, 1)$ , so its inverse

$$G : (0, 1) \longrightarrow (0, 1), \quad F(G(y)) = y, \quad (2.3)$$

is well defined. Reflection gives  $G(1 - y) = 1 - G(y)$ ; hence only  $y \downarrow 0$  needs to be studied.

The Laplace transform and its logarithm are

$$M(t) := \mathbb{E}e^{-tX} = \prod_{j \geq 1} \frac{1 - e^{-t/2^j}}{t/2^j}, \quad (2.4)$$

$$K(t) := \log M(t) = \sum_{j \geq 1} \log \left( \frac{1 - e^{-t/2^j}}{t/2^j} \right). \quad (2.5)$$

The Bromwich representation

$$F(x) = \frac{1}{2\pi i} \int_{\sigma - i\infty}^{\sigma + i\infty} \frac{\exp(xt + K(t))}{t} dt, \quad \sigma > 0, \quad (2.6)$$

is the analytic source of the saddle expansion.

### 2.2 Exact Lambert phase and periodic data

Set throughout

$$L := \log 2, \quad B_0 := 1 + \frac{L}{2}, \quad \beta := \frac{B_0}{L} = \frac{1}{L} + \frac{1}{2}. \quad (2.7)$$

For sufficiently small  $x > 0$ , define the exact lower-Lambert phase  $\lambda$  by

$$x = \lambda e^{-L\lambda}, \quad \lambda(x) = -\frac{1}{L} W_{-1}(-Lx). \quad (2.8)$$

The imported forward theorem states that, for every fixed  $N \geq 1$ ,

$$\log F(x) = -\frac{L}{2} \lambda^2 + B_0 \lambda - \frac{1}{2} \log \lambda + C_{\sharp} + \Psi(\lambda) + \sum_{j=1}^{N-1} \frac{A_j(\lambda)}{\lambda^j} + \mathcal{O}_N(\lambda^{-N}). \quad (2.9)$$

Every  $A_j$  and  $\Psi$  is smooth, bounded, and one-periodic, and the remainder is uniform in the phase. With Euler's constant  $\gamma$  and the first Stieltjes constant  $\gamma_1$ ,

$$C_{\#} = \frac{6\gamma^2 + 12\gamma_1 - \pi^2}{12L} - \frac{7L}{12} - \frac{1}{2} \log \pi. \quad (2.10)$$

The centered periodic fluctuation has Fourier expansion

$$\Psi(u) = \sum_{k \neq 0} \widehat{\Psi}(k) e^{2\pi i k u}, \quad \widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k) \zeta(1 - \chi_k)}{L}, \quad \chi_k = \frac{2\pi i k}{L}. \quad (2.11)$$

The first forward correction is already explicit in the corpus:

$$A_1(u) = \frac{1}{24} + \frac{1}{2L} - \frac{\Psi''(u) + \Psi'(u)^2}{2L^2}. \quad (2.12)$$

The Gamma factor gives exponentially decaying Fourier coefficients, so all phase differentiations below are harmless.

### 2.3 Leading inverse scale

For  $y \downarrow 0$  put

$$T := \log \frac{1}{y}, \quad \rho := \sqrt{\frac{2T}{L}}, \quad z := \rho^{-1}, \quad \ell := \log \rho, \quad \vartheta := \rho + \beta. \quad (2.13)$$

The leading inverse equivalent is

$$G_0(y) := \rho e^{-L\rho - L\beta} = \frac{\sqrt{T}}{e\sqrt{L}} e^{-\sqrt{2LT}}. \quad (2.14)$$

The exact phase of  $G(y)$  is written

$$\lambda(G(y)) = \rho + \beta + D(z), \quad D(z) = \sum_{n \geq 1} d_n z^n. \quad (2.15)$$

Here and below  $\ell$  and  $\vartheta$  are treated as independent coefficient variables during formal coefficient extraction. Only after extraction do we substitute  $\ell = \log \rho$  and  $\vartheta = \rho + \beta$ . This is the standard two-scale convention; expanding the fast phase itself as an ordinary slow series would destroy terms at their natural order.

## 3 Closed all-orders phase reversion

### 3.1 The residual functional

Define the formal residual

$$\begin{aligned} \mathcal{R}(z, u; \ell, \vartheta) := & -\frac{L}{2}\beta^2 + \frac{L}{2}u^2 + \frac{1}{2}\ell + \frac{1}{2}\log(1 + \beta z + zu) - C_{\#} - \Psi(\vartheta + u) \\ & - \sum_{j \geq 1} A_j(\vartheta + u) z^j (1 + \beta z + zu)^{-j}. \end{aligned} \quad (3.1)$$

Substitution of (2.15) into the forward expansion gives the formal phase equation

$$0 = \frac{L}{z} D(z) + \mathcal{R}(z, D(z); \ell, \vartheta), \quad (3.2)$$

or equivalently

$$D(z) = z\Phi(z, D(z)), \quad \Phi(z, u) := -\frac{1}{L}\mathcal{R}(z, u). \quad (3.3)$$

The repository obtains  $d_n$  by a triangular recursion. The next theorem turns that recursion into one finite coefficient extraction.

**Theorem 3.1** (Closed generalized Lagrange–Bürmann formula). *For every  $n \geq 1$ , the coefficient in (2.15) is*

$$d_n = \sum_{k=1}^n \frac{1}{k} [z^{n-k} u^{k-1}] \Phi(z, u)^k = \sum_{k=1}^n \frac{(-1)^k}{k L^k} [z^{n-k} u^{k-1}] \mathcal{R}(z, u)^k. \quad (3.4)$$

At order  $n$  the formula uses only

$$\Psi, \Psi', \dots, \Psi^{(n-1)}, \quad A_1, \dots, A_{n-1},$$

and finitely many derivatives of those  $A_j$ . Thus (3.4) is a finite formula at every prescribed order despite the infinite sum in (3.1).

*Proof.* Introduce an auxiliary variable  $t$  and let  $W(t, z)$  be the unique formal solution of

$$W = t\Phi(z, W).$$

For fixed  $z$ , ordinary Lagrange inversion in  $t$  gives

$$W(t, z) = \sum_{k \geq 1} \frac{t^k}{k} [u^{k-1}] \Phi(z, u)^k.$$

Since  $D(z) = W(z, z)$ , taking  $[z^n]$  after setting  $t = z$  yields the first formula in (3.4); the second follows from  $\Phi = -\mathcal{R}/L$ .

In a term contributing to  $d_n$ , the explicit  $z$ -degree is at most  $n - 1$ . Consequently  $A_j z^j$  cannot occur for  $j \geq n$ . The extraction  $[u^{k-1}]$  requires at most  $n - 1$  phase derivatives. All operations are therefore finite.  $\square$

*Remark 3.2* (Why this is more than a restatement of the recursion). The triangular recursion computes  $d_1, d_2, \dots$  sequentially. Formula (3.4) expresses  $d_n$  directly as a finite sum over  $k$ , with no reference to previously computed coefficients. Combinatorially, it is a sum over rooted substitution trees collapsed into one coefficient extractor. This form is better suited to symbolic generation, degree estimates, and Lean formalization through a general Lagrange-inversion theorem.

### 3.2 The first three phase coefficients

All periodic functions in the next formulas are evaluated at  $\vartheta$ . The first two coefficients agree with the existing inverse report:

$$d_1 = \frac{\beta^2}{2} + \frac{C_\# + \Psi - \frac{1}{2}\ell}{L}, \quad (3.5)$$

$$d_2 = \frac{\Psi' d_1 + A_1 - \frac{1}{2}\beta}{L}. \quad (3.6)$$

The closed extractor gives the next term without further inversion.

**Proposition 3.3** (Explicit third phase correction). *The coefficient of  $\rho^{-3}$  in the inverse Lambert phase is*

$$d_3 = \frac{1}{L} \left( \Psi' d_2 + \frac{1}{2} \Psi'' d_1^2 + A_1' d_1 - \beta A_1 + A_2 - \frac{L}{2} d_1^2 - \frac{1}{2} d_1 + \frac{\beta^2}{4} \right). \quad (3.7)$$

*Proof.* Take  $[z^2]$  in (3.2). The contributions are respectively

$$\frac{L}{2} d_1^2, \quad \frac{1}{2} d_1 - \frac{1}{4} \beta^2, \quad -\Psi' d_2 - \frac{1}{2} \Psi'' d_1^2, \quad -A_1' d_1 + \beta A_1, \quad -A_2.$$

Their sum plus  $L d_3$  vanishes, giving (3.7).  $\square$

## 4 Direct formulas for the inverse coefficients

### 4.1 Logarithmic and multiplicative forms

Since  $G(y) = \lambda e^{-L\lambda}$  and  $G_0(y) = \rho e^{-L(\rho+\beta)}$ , (2.15) gives the exact formal identity

$$\frac{G(y)}{G_0(y)} = (1 + \beta z + zD(z))e^{-LD(z)}. \quad (4.1)$$

Define logarithmic and multiplicative coefficients by

$$\log \frac{G(y)}{G_0(y)} \sim \sum_{n \geq 1} h_n z^n, \quad (4.2)$$

$$\frac{G(y)}{G_0(y)} \sim 1 + \sum_{n \geq 1} g_n z^n. \quad (4.3)$$

Then

$$\sum_{n \geq 1} h_n z^n = \log(1 + \beta z + zD) - LD. \quad (4.4)$$

The existing corpus records the complete exponential-Bell conversion  $h \mapsto g$ . We now give a direct  $d \mapsto g$  formula.

Let  $\mathcal{B}_{n,k}$  denote the exponential partial Bell polynomial, normalized by

$$\left( \sum_{j \geq 1} d_j z^j \right)^k = \sum_{n \geq k} \frac{k!}{n!} \mathcal{B}_{n,k}(1!d_1, 2!d_2, \dots) z^n. \quad (4.5)$$

Use  $\mathcal{B}_{0,0} = 1$  and  $\mathcal{B}_{n,0} = 0$  for  $n > 0$ .

**Theorem 4.1** (Direct partial-Bell formula for  $g_n$ ). *For every  $n \geq 1$ , set  $x_j = j!d_j$ . Then*

$$\begin{aligned} g_n &= \frac{1}{n!} \sum_{k=1}^n (-L)^k \mathcal{B}_{n,k}(x_1, x_2, \dots) \\ &\quad + \frac{\beta}{(n-1)!} \sum_{k=0}^{n-1} (-L)^k \mathcal{B}_{n-1,k}(x_1, x_2, \dots) \\ &\quad + \frac{1}{(n-1)!} \sum_{k=1}^{n-1} k(-L)^{k-1} \mathcal{B}_{n-1,k}(x_1, x_2, \dots). \end{aligned} \quad (4.6)$$

Combining (3.4) and (4.6) is a nonrecursive finite formula for the arbitrary-order inverse coefficient  $g_n$ .

*Proof.* Expand the factorized expression (4.1). The coefficient of  $z^n$  in  $e^{-LD}$  is

$$\frac{1}{n!} \sum_{k=1}^n (-L)^k \mathcal{B}_{n,k}(x_1, x_2, \dots).$$

Multiplication by  $\beta z$  gives the second line of (4.6). Finally,

$$[z^m] D e^{-LD} = \frac{1}{m!} \sum_{k=1}^m k(-L)^{k-1} \mathcal{B}_{m,k}(x_1, x_2, \dots),$$

which gives the third line with  $m = n - 1$ . □

For comparison, the logarithmic-to-multiplicative conversion is

$$g_n = \frac{1}{n!} \mathcal{B}_n(1!h_1, 2!h_2, \dots, n!h_n), \quad (4.7)$$

where  $\mathcal{B}_n$  is the complete exponential Bell polynomial.

## 4.2 Explicit coefficients through third order

Equations (4.4) and (4.1) give

$$h_1 = \beta - Ld_1, \quad (4.8)$$

$$h_2 = d_1 - \frac{\beta^2}{2} - Ld_2, \quad (4.9)$$

$$h_3 = d_2 - \beta d_1 + \frac{\beta^3}{3} - Ld_3. \quad (4.10)$$

In particular,

$$h_1 = \frac{1}{2}\ell + \kappa_0 - \Psi(\vartheta), \quad \kappa_0 := \beta - \frac{L}{2}\beta^2 - C_\sharp, \quad (4.11)$$

which is the first oscillatory relative correction already present in the corpus. The multiplicative coefficients are

$$g_1 = h_1, \quad (4.12)$$

$$g_2 = h_2 + \frac{1}{2}h_1^2, \quad (4.13)$$

$$g_3 = h_3 + h_1h_2 + \frac{1}{6}h_1^3. \quad (4.14)$$

Directly in terms of the  $d_j$ ,

$$g_2 = d_1 - Ld_2 - \beta Ld_1 + \frac{L^2}{2}d_1^2, \quad (4.15)$$

$$g_3 = -Ld_3 + (1 - \beta L)d_2 + L^2d_1d_2 + \left(\frac{\beta L^2}{2} - L\right)d_1^2 - \frac{L^3}{6}d_1^3. \quad (4.16)$$

Substitution of (3.5)–(3.7) makes  $g_3$  an explicit polynomial in  $\ell$  with periodic coefficients generated by  $\Psi, A_1, A_2$  and their derivatives.

## 4.3 Universal highest slow logarithm

**Theorem 4.2** (Universal leading logarithm at every order). *For  $n \geq 1$ ,  $g_n(\ell, \vartheta)$  is a polynomial in  $\ell$  of degree at most  $n$ , and*

$$\boxed{[\ell^n]g_n(\ell, \vartheta) = \frac{1}{2^n n!}}. \quad (4.17)$$

Equivalently,

$$g_n(\ell, \vartheta) = \frac{\ell^n}{2^n n!} + \mathcal{O}_n((1 + |\ell|)^{n-1}), \quad (4.18)$$

uniformly in  $\vartheta$ . The top logarithm is independent of every periodic function  $\Psi, A_1, A_2, \dots$

*Proof.* Write  $\mathcal{R} = \ell/2 + \mathcal{R}_0$ , where  $\mathcal{R}_0$  is independent of  $\ell$ . Formula (3.4) immediately gives

$$d_1 = -\frac{\ell}{2L} + \mathcal{O}(1).$$

For  $n \geq 2$ , a term contributing to  $[z^{n-k}u^{k-1}]\mathcal{R}^k$  cannot use the explicit  $\ell/2$  from all  $k$  factors, because then its  $u$ -degree would be zero. Hence

$$\deg_\ell d_n \leq n - 1 \quad (n \geq 2). \quad (4.19)$$

Now use (4.1). At order  $z^n$ , a contribution of  $\ell$ -degree  $n$  can only come from the term  $(-LD)^n/n!$  in  $e^{-LD}$ , using  $d_1z$  from every copy of  $D$ . Its coefficient is

$$\frac{(-L)^n}{n!} \left(-\frac{1}{2L}\right)^n = \frac{1}{2^n n!}.$$

The prefactor  $1 + \beta z + zD$  and every occurrence of  $d_j$  with  $j \geq 2$  reduce the maximum logarithmic degree by at least one.  $\square$

*Remark 4.3.* The theorem explains why a purely elementary approximation can look surprisingly accurate: the largest slow logarithms exponentiate as though the periodic term were absent. The periodic data enter one logarithmic degree lower, but they remain indispensable because their size is the same as the other coefficients at that degree.

#### 4.4 Rigorous arbitrary-order expansion

**Theorem 4.4** (Complete phase-aware inverse expansion). *Let the coefficients  $d_n$  be defined by (3.4) and  $g_n$  by (4.6), with  $g_0 = 1$ . For every fixed  $N \geq 1$ ,*

$$G(y) = G_0(y) \left( \sum_{n=0}^{N-1} \frac{g_n(\log \rho, \rho + \beta)}{\rho^n} + \mathcal{O}_N \left( \frac{(1 + \log \rho)^N}{\rho^N} \right) \right), \quad y \downarrow 0. \quad (4.20)$$

The error is uniform in the phase  $\rho + \beta \pmod{1}$ . The analogous right-endpoint expansion is

$$1 - G(1 - y) = G(y), \quad (4.21)$$

so the same coefficients govern both ends.

*Proof.* The imported forward theorem (2.9) is uniform in the phase at every finite order. Substitution of  $\lambda = \rho + \beta + D$  produces (3.2); its derivative with respect to  $D$  is  $L/z + \mathcal{O}(1) = L\rho + \mathcal{O}(1)$ . The formal solution is therefore asymptotically stable under one Newton correction. Truncating  $D$  after  $d_{N-1}z^{N-1}$  leaves a residual of order  $\mathcal{O}_N((1 + \ell)^{N-1}z^{N-1})$ , and division by the derivative gives a phase error  $\mathcal{O}_N((1 + \ell)^{N-1}z^N)$ . The analytic map  $\lambda \mapsto \lambda e^{-L\lambda}$  transfers this to a relative inverse error of the same order. Expanding the exponential in (4.1) adds at most one logarithmic degree, yielding the remainder in (4.20). Reflection of  $F$  proves the right-endpoint statement.  $\square$

### 5 All-orders inverse derivatives

The two-term elasticity is already known in the corpus. The coefficient calculus above gives all orders with almost no additional work.

#### 5.1 Two-scale differential operator

Because  $T = L\rho^2/2$  and  $y = e^{-T}$ ,

$$y \frac{d}{dy} = -\frac{d}{dT} = -\frac{1}{L\rho} \frac{d}{d\rho}. \quad (5.1)$$

When a coefficient is written as  $a(\ell, \vartheta)\rho^{-n}$ , total  $\rho$ -differentiation is

$$\mathcal{D}_\rho(a(\ell, \vartheta)\rho^{-n}) = \rho^{-n} \partial_\vartheta a + \rho^{-n-1} (\partial_\ell a - na). \quad (5.2)$$

Thus repeated logarithmic  $y$ -derivatives are generated by the operator

$$\mathcal{Y} := -\frac{1}{L\rho} \mathcal{D}_\rho. \quad (5.3)$$

**Theorem 5.1** (Arbitrary-order quantile elasticity). *Let*

$$\frac{yG'(y)}{G(y)} \sim \sum_{m \geq 1} \frac{e_m(\ell, \vartheta)}{\rho^m}. \quad (5.4)$$

Then

$$e_1 = 1, \quad (5.5)$$

$$e_2 = -\frac{1 + \partial_{\vartheta} h_1}{L}, \quad (5.6)$$

$$e_m = -\frac{1}{L} (\partial_{\vartheta} h_{m-1} + \partial_{\ell} h_{m-2} - (m-2)h_{m-2}), \quad m \geq 3. \quad (5.7)$$

In particular,

$$\frac{yG'(y)}{G(y)} = \frac{1}{\rho} + \frac{\Psi'(\vartheta) - 1}{L\rho^2} + \mathcal{O}\left(\frac{1 + \log \rho}{\rho^3}\right), \quad (5.8)$$

recovering the existing two-term theorem.

*Proof.* From (4.2),

$$\log G = \log \rho - L\rho - L\beta + \sum_{n \geq 1} h_n \rho^{-n}.$$

Apply  $-(L\rho)^{-1}\mathcal{D}_{\rho}$  and collect equal powers of  $\rho^{-1}$ . The derivative of the leading part contributes  $\rho^{-1} - (L\rho^2)^{-1}$ . Formula (5.2) gives the remaining terms and hence (5.5)–(5.7). Since  $\partial_{\vartheta} h_1 = -\Psi'$ , (5.8) follows.  $\square$

**Corollary 5.2** (Higher logarithmic and ordinary derivatives). *For every fixed  $r \geq 1$ , the complete asymptotic expansion of  $(y \, d/dy)^r \log G(y)$  is obtained by applying  $\mathcal{Y}^r$  to*

$$\log \rho - L\rho - L\beta + \sum_{n \geq 1} h_n(\ell, \vartheta) \rho^{-n}.$$

More explicitly, put  $E := y \, d/dy$ ,  $f := \log G$ , and let  $\mathfrak{s}(k, j)$  be the signed Stirling number of the first kind. Then

$$H_k := y^k f^{(k)}(y) = \sum_{j=1}^k \mathfrak{s}(k, j) E^j f(y), \quad (5.9)$$

and weighted homogeneity of the complete exponential Bell polynomial gives

$$\boxed{\frac{y^r G^{(r)}(y)}{G(y)} = \mathcal{B}_r(H_1, H_2, \dots, H_r).} \quad (5.10)$$

Consequently, (5.9)–(5.10), together with the two-scale operator (5.3), constitute a finite coefficient algorithm for every ordinary derivative of the inverse quantile.

## 6 A constrained-partition formula for the forward saddle coefficients

The inverse formulas require the periodic forward coefficients  $A_j$ . The repository constructs them through a saddle coefficient engine. This section gives a closed finite partition sum for the universal local Gaussian part of that engine.

### 6.1 Normalized saddle jets

Let  $s > 0$  solve the saddle equation

$$x = -K'(s). \quad (6.1)$$

Define normalized derivatives

$$\beta_r := s^r K^{(r)}(s), \quad \rho_r := \frac{\beta_r}{\beta_2}, \quad r \geq 3. \quad (6.2)$$

Scaling the Bromwich contour by  $t = s(1 + iZ/\sqrt{\beta_2})$  yields the formal correction factor

$$\mathcal{S} = \mathbb{E} \left[ \left( 1 + \frac{iZ}{\sqrt{\beta_2}} \right)^{-1} \exp \left( \sum_{r \geq 3} \frac{\rho_r}{r!} \frac{(iZ)^r}{\beta_2^{(r-2)/2}} \right) \right], \quad Z \sim N(0, 1). \quad (6.3)$$

The first factor is essential: it comes from the  $1/t$  in the CDF Bromwich integral. Deleting it would produce the density coefficients instead. Write

$$\mathcal{S} \sim 1 + \sum_{n \geq 1} \frac{C_n}{\beta_2^n}. \quad (6.4)$$

**Theorem 6.1** (Finite constrained-partition formula). *For  $n \geq 1$ ,*

$$C_n = \sum_{\substack{m_0, m_3, m_4, \dots \geq 0 \\ m_0 + \sum_{r \geq 3} (r-2)m_r = 2n}} (-1)^{m_0 + N/2} (N-1)!! \prod_{r \geq 3} \frac{\rho_r^{m_r}}{m_r! (r!)^{m_r}}, \quad (6.5)$$

where

$$N := m_0 + \sum_{r \geq 3} r m_r. \quad (6.6)$$

The constraint forces  $N$  to be even. Only  $3 \leq r \leq 2n+2$  can occur, so (6.5) is finite.

*Proof.* Expand the denominator geometrically, taking  $m_0$  factors of  $-iZ/\sqrt{\beta_2}$ , and expand the exponential by taking  $m_r$  copies of the  $r$ th saddle monomial. The total inverse half-power of  $\beta_2$  is

$$\frac{1}{2} \left( m_0 + \sum_{r \geq 3} (r-2)m_r \right),$$

which equals  $n$  precisely under the displayed constraint. The total Gaussian degree is  $N$ ; its parity agrees with the constraint because  $N - 2n = 2 \sum_{r \geq 3} m_r$ . Therefore  $\mathbb{E} Z^N = (N-1)!!$ , and the powers of  $i$  together with the denominator sign give  $(-1)^{m_0 + N/2}$ . Multiplicity factorials arise from the exponential series.  $\square$

The first two coefficients are

$$C_1 = -1 - \frac{1}{2}\rho_3 + \frac{1}{8}\rho_4 - \frac{5}{24}\rho_3^2, \quad (6.7)$$

$$C_2 = 3 + \frac{5}{2}\rho_3 - \frac{5}{8}\rho_4 + \frac{1}{8}\rho_5 - \frac{1}{48}\rho_6 + \frac{35}{24}\rho_3^2 - \frac{35}{48}\rho_3\rho_4 + \frac{7}{48}\rho_3\rho_5 \\ + \frac{35}{384}\rho_4^2 + \frac{35}{48}\rho_3^3 - \frac{35}{64}\rho_3^2\rho_4 + \frac{385}{1152}\rho_3^4. \quad (6.8)$$

When every  $\rho_r$  vanishes, (6.5) reduces to

$$C_n = (-1)^n (2n-1)!!, \quad (6.9)$$

the asymptotic coefficients of the Gaussian Mills ratio. This is a sensitive sign and normalization check.

## 6.2 Logarithmic saddle coefficients

Writing

$$\log \mathcal{S} \sim \sum_{n \geq 1} \frac{\Lambda_n}{\beta_2^n}, \quad (6.10)$$

we obtain the nonrecursive composition formula

$$\Lambda_n = \sum_{k=1}^n \frac{(-1)^{k-1}}{k} \sum_{\substack{n_1 + \dots + n_k = n \\ n_j \geq 1}} C_{n_1} \cdots C_{n_k}. \quad (6.11)$$

Equivalently,  $\Lambda_n$  is an ordinary partial-Bell polynomial in  $C_1, \dots, C_n$ . The first two are

$$\Lambda_1 = C_1, \quad (6.12)$$

$$\Lambda_2 = C_2 - \frac{1}{2}C_1^2. \quad (6.13)$$

The accompanying Python program generates  $C_n$  and  $\Lambda_n$  at arbitrary fixed order by enumerating the finite constraint in (6.5).

### 6.3 Connection to the periodic forward coefficients

Let  $q = \log_2 s$ ,  $D_q = d/dq$ , and let  $J(q) = K(2^q)$ . The normalized ordinary derivatives are generated exactly by

$$\beta_r(q) = L^{-r} \prod_{j=0}^{r-1} (D_q - jL)J(q). \quad (6.14)$$

After inserting the quadratic-periodic normal form for  $J$  from [section 7](#), formulas (6.5) and (6.11), followed by the finite conversion from saddle  $q$  to exact Lambert  $\lambda$ , generate every forward  $A_j$ . Thus the full coefficient pipeline is

$$\Psi \xrightarrow{(6.14)} \{\beta_r\} \xrightarrow{(6.5)} \{C_n\} \xrightarrow{(6.11)} \{\Lambda_n\} \longrightarrow \{A_j\} \xrightarrow{(3.4)} \{d_n\} \xrightarrow{(4.6)} \{g_n\}. \quad (6.15)$$

Every arrow is a finite differential-polynomial or coefficient-extraction operation at fixed order.

## 7 An exact 2-adic completion of the Laplace exponent

This section develops the second main new structure. It is exact, not merely asymptotic.

### 7.1 Dyadic difference equation

Let

$$g(t) := \log\left(\frac{1 - e^{-t}}{t}\right). \quad (7.1)$$

The product definition gives

$$K(2t) - K(t) = g(t) = \log(1 - e^{-t}) - \log t. \quad (7.2)$$

With  $q = \log_2 t$ , the quadratic polynomial

$$Q(q) := -\frac{L}{2}q^2 + \frac{L}{2}q \quad (7.3)$$

satisfies  $Q(q+1) - Q(q) = -Lq$ . The remaining nonperiodic defect is removed by the following positive function.

**Definition 7.1** (Dyadic Bose tail). For  $t > 0$ , define

$$B(t) := -\sum_{k \geq 0} \log(1 - e^{-2^k t}). \quad (7.4)$$

Then

$$B(t) = -\log(1 - e^{-t}) + B(2t). \quad (7.5)$$

The cubic-pole constant used below is

$$C_B := \frac{L}{12} + \frac{\pi^2 - 6\gamma^2 - 12\gamma_1}{12L}. \quad (7.6)$$

It is related to the forward saddle constant by the compact identity

$$C_{\sharp} + C_B = -\frac{1}{2} \log(2\pi). \quad (7.7)$$

**Theorem 7.2** (Exact quadratic–periodic–dyadic decomposition). *For every  $t > 0$ ,*

$$K(t) = -\frac{\log^2 t}{2L} + \frac{1}{2} \log t - C_B + \Psi(\log_2 t) + B(t). \quad (7.8)$$

Equivalently,

$$K(2^q) = Q(q) + \Psi(q) - C_B + B(2^q). \quad (7.9)$$

*Proof.* Set

$$\Omega(q) := K(2^q) - Q(q) - B(2^q).$$

By (7.2), (7.5), and the defining difference of  $Q$ , one has  $\Omega(q+1) = \Omega(q)$ . The Mellin expansion of  $B(t)$  as  $t \downarrow 0$ , proved in theorem 7.5, is

$$B(t) = \frac{\log^2(1/t)}{2L} + \frac{1}{2} \log(1/t) + C_B - \Psi(\log_2 t) + o(1).$$

Since  $K(t) \rightarrow 0$  as  $t \downarrow 0$ , taking  $q = m + u \rightarrow -\infty$  with fixed  $u \in [0, 1)$  gives  $\Omega(u) = \Psi(u) - C_B$ . Periodicity upgrades this limiting identity to every real  $q$ , proving (7.9).  $\square$

## 7.2 Collapse to one 2-adically weighted exponential series

**Theorem 7.3** (Exact 2-adic coefficient formula). *For every  $t > 0$ ,*

$$B(t) = \sum_{n \geq 1} b_n e^{-nt}, \quad b_n := \frac{2^{\nu_2(n)+1} - 1}{n}. \quad (7.10)$$

The coefficients satisfy

$$0 < b_n < 2, \quad nb_n = 2^{\nu_2(n)+1} - 1. \quad (7.11)$$

Consequently, for every  $N \geq 0$ ,

$$0 < B(t) - \sum_{n=1}^N b_n e^{-nt} \leq \frac{2e^{-(N+1)t}}{1 - e^{-t}}. \quad (7.12)$$

*Proof.* Expand each Bose logarithm:

$$B(t) = \sum_{k \geq 0} \sum_{m \geq 1} \frac{1}{m} e^{-m2^k t}.$$

For a fixed  $n$ , the representations  $n = m2^k$  are indexed by  $0 \leq k \leq \nu_2(n)$ , and their total coefficient is

$$\sum_{k=0}^{\nu_2(n)} \frac{1}{n/2^k} = \frac{1}{n} \sum_{k=0}^{\nu_2(n)} 2^k = \frac{2^{\nu_2(n)+1} - 1}{n}.$$

If  $n = 2^v m$  with  $m$  odd, then  $b_n = (2 - 2^{-v})/m < 2$ . Summing the geometric majorant proves (7.12).  $\square$

The first terms are

$$B(t) = e^{-t} + \frac{3}{2}e^{-2t} + \frac{1}{3}e^{-3t} + \frac{7}{4}e^{-4t} + \frac{1}{5}e^{-5t} + \frac{1}{2}e^{-6t} + \dots \quad (7.13)$$

The spikes near powers of two are not noise: they are the exact arithmetic trace of the dyadic scale hierarchy.

### 7.3 Mahler and Dirichlet forms

Put

$$\mathcal{B}(z) := \sum_{n \geq 1} b_n z^n, \quad |z| < 1. \quad (7.14)$$

Then

$$\mathcal{B}(z) = - \sum_{k \geq 0} \log(1 - z^{2^k}) = -\log(1 - z) + \mathcal{B}(z^2). \quad (7.15)$$

Thus  $\mathcal{B}$  is a logarithmic Mahler function. Its coefficients have the Dirichlet series

**Theorem 7.4** (Dirichlet series of the dyadic Bose coefficients). *For  $\Re s > 0$ ,*

$$\boxed{\sum_{n \geq 1} \frac{b_n}{n^s} = \frac{\zeta(s+1)}{1 - 2^{-s}}.} \quad (7.16)$$

*Proof.* Use the original double indexing:

$$\sum_{n \geq 1} \frac{b_n}{n^s} = \sum_{k \geq 0} \sum_{m \geq 1} \frac{1}{m} (m2^k)^{-s} = \zeta(s+1) \sum_{k \geq 0} 2^{-ks}.$$

Absolute convergence holds for  $\Re s > 0$ . □

### 7.4 One Mellin kernel, two asymptotic ends

**Theorem 7.5** (Mellin bridge). *For  $\Re s > 0$ ,*

$$\int_0^\infty B(t) t^{s-1} dt = \Gamma(s) \frac{\zeta(s+1)}{1 - 2^{-s}}. \quad (7.17)$$

Hence, for any  $c > 0$ ,

$$B(t) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \Gamma(s) \frac{\zeta(s+1)}{1 - 2^{-s}} t^{-s} ds. \quad (7.18)$$

As  $t \downarrow 0$ , for every fixed  $M \geq 1$ ,

$$B(t) = \frac{R^2}{2L} + \frac{R}{2} + C_B - \Psi(\log_2 t) + \sum_{m=1}^{M-1} c_m t^m + \mathcal{O}_M(t^M), \quad R := \log(1/t), \quad (7.19)$$

$$c_m = \frac{(-1)^m \zeta(1-m)}{m!(1-2^m)}. \quad (7.20)$$

As  $t \rightarrow \infty$ , the exact expansion is the convergent 2-adic exponential series (7.10).

*Proof.* Termwise Mellin transformation of (7.10) gives (7.17); Mellin inversion gives (7.18). Shift the contour left. At  $s = 0$ , the factors  $\Gamma(s)$ ,  $\zeta(1+s)$ , and  $(1 - e^{-Ls})^{-1}$  contribute an order-three pole. Their product begins

$$\frac{1}{Ls^3} + \frac{1}{2s^2} + \left( \frac{L}{12} + \frac{\pi^2/12 - \gamma_1 - \gamma^2/2}{L} \right) \frac{1}{s} + \mathcal{O}(1),$$

which yields the quadratic polynomial and  $C_B$ . The nonzero poles  $s = \chi_k = 2\pi i k / L$  have residues

$$\frac{\Gamma(\chi_k) \zeta(1 + \chi_k)}{L} t^{-\chi_k}.$$

After reindexing  $k \mapsto -k$ , their sum is exactly  $-\Psi(\log_2 t)$  by (2.11). Finally, the pole of  $\Gamma$  at  $s = -m$  contributes  $c_m t^m$ . Standard vertical-line decay of  $\Gamma$  controls the shifted contour. □

The periodic correction and the 2-adic exponential sectors are not two unrelated phenomena. They are the small- $t$  and large- $t$  residue/Dirichlet faces of the single Mellin kernel

$$\Gamma(s)\zeta(s+1)/(1-2^{-s}).$$

The poles  $2\pi i k/L$  generate the Gamma-zeta phase; regrouping the same kernel as a Dirichlet series produces the exact weights  $(2^{\nu_2(n)+1} - 1)/n$ .

## 7.5 Exact saddle map and differential jets

Differentiating (7.8) gives an exact correction to the saddle equation.

**Corollary 7.6** (Exact 2-adic saddle map). *Let  $t = 2^q$ . The saddle abscissa  $x = -K'(t)$  is*

$$x(q) = 2^{-q} \left( q - \frac{1}{2} - \frac{\Psi'(q)}{L} \right) + \sum_{n \geq 1} (2^{\nu_2(n)+1} - 1) e^{-n2^q}. \quad (7.21)$$

The first term is the complete algebraic periodic saddle map; the second is an exact, convergent, ultraflat dyadic correction.

*Proof.* Differentiate (7.8). Since  $B'(t) = -\sum n b_n e^{-nt}$  and  $n b_n = 2^{\nu_2(n)+1} - 1$ , negation gives (7.21).  $\square$

For derivative bookkeeping, let  $T_r$  be the Touchard polynomial

$$T_r(x) = \sum_{j=0}^r \left\{ \begin{matrix} r \\ j \end{matrix} \right\} x^j, \quad (7.22)$$

where braces denote Stirling numbers of the second kind. Then

$$\frac{d^r}{dq^r} B(2^q) = L^r \sum_{n \geq 1} b_n T_r(-n2^q) e^{-n2^q}. \quad (7.23)$$

Ordinary  $t$ -derivatives are even simpler:

$$t^r B^{(r)}(t) = (-1)^r \sum_{n \geq 1} b_n (nt)^r e^{-nt}. \quad (7.24)$$

These identities give every exponentially small correction to the normalized saddle jets  $\beta_r$ . They also explain a technical obstacle to naive hyperasymptotics: phase derivatives of an  $e^{-t}$  sector acquire powers of  $t$ , so the ordinary inverse-power saddle hierarchy is not uniform inside a fixed dyadic exponential sector.

## 8 A practical arbitrary-order algorithm

**Algorithm 8.1** (Closed coefficient generation). *Fix an output order  $N$ .*

1. Represent  $\Psi$  and  $A_1, \dots, A_{N-1}$  by exact symbolic expressions, Fourier series, or differentiable coefficient oracles. Formula (2.12) supplies  $A_1$  directly; higher  $A_j$  can be generated from the saddle pipeline (6.15).
2. Form the truncated residual  $\mathcal{R}_N(z, u)$  by discarding explicit powers  $z^j$  with  $j \geq N$  and Taylor-expanding in  $u$  through degree  $N - 1$ .
3. For each  $1 \leq n \leq N$ , compute  $d_n$  directly from (3.4). No earlier  $d_j$  is mathematically required, although reusing intermediate products is computationally advantageous.

4. Compute  $g_n$  from (4.6), or compute  $h_n$  from (4.4) and use (4.7).
5. Evaluate at  $\ell = \log \rho$ ,  $\vartheta = \rho + \beta$  and assemble (4.20).
6. For derivatives, use (5.5)–(5.7) or repeatedly apply  $\mathcal{V}$  from (5.3).

## 8.1 Finite complexity at fixed order

The sum in (3.4) has only  $N$  outer terms. Its cost is governed by bivariate truncated multiplication through degrees  $(N, N)$  and by phase-jet composition. With dense polynomial arithmetic, a straightforward implementation is polynomial in the number of retained monomials; sparse Fourier arithmetic is usually much cheaper because the Gamma factor makes high modes negligible. The constrained saddle formula (6.5) enumerates partitions of the weight  $2n$ , so its raw term count grows subexponentially like an integer-partition function rather than like all set partitions.

## 8.2 Numerical stability

Three rules are important in actual computation.

- Preserve the exact phase  $\vartheta = \rho + \beta$  inside every periodic coefficient. Replacing it by  $\rho$  changes the first oscillatory correction at order  $\rho^{-1}$ .
- Evaluate  $G$  through its logarithm when  $y$  is extremely small:

$$\log G = \log G_0 + \sum_{n < N} h_n \rho^{-n}.$$

This avoids underflow and loss of relative precision.

- Use the Fourier form (2.11) for  $\Psi$  and its derivatives. Differentiation multiplies the  $k$ th mode by  $2\pi i k$ , while the Gamma factor still supplies exponential decay.

# 9 Symbolic and numerical verification

The archive contains the fully commented program `inverse_fabius_experiments.py`. It independently implements the constrained-partition formula, the generalized Lagrange extractor through third order, the exact 2-adic series, the Dirichlet identity, the Mellin residue expansion, and the exact decomposition of  $K(t)$ .

## 9.1 Symbolic checks

SymPy returned identically zero for

$$d_j^{\text{closed Lagrange}} - d_j^{\text{direct coefficient recursion}}, \quad j = 1, 2, 3. \quad (9.1)$$

The generated  $C_1, C_2, C_3$  specialize at  $\rho_3 = \rho_4 = \dots = 0$  to  $-1, 3, -15$ , matching (6.9). The machine-generated expressions are retained in `generated_coefficients.tex` and the plain-text log in `verification_output.txt`.

## 9.2 High-precision checks

At 70 decimal digits, using twelve positive Fourier modes, the exact identity (7.8) agreed with the independent defining product for  $K(t)$  to approximately  $1.7 \times 10^{-16}$  or better at  $t = 0.1, 1, 3, 10$ . The product and the 2-adic exponential series for  $B(t)$  agreed to  $7.5 \times 10^{-39}$  at  $t = 0.7$  and to  $3.8 \times 10^{-55}$  at  $t = 1.5$  with the finite truncations recorded in the script. The small- $t$  Mellin-residue expansion through

ten algebraic residues and twelve Fourier modes agreed with the product to about  $4.1 \times 10^{-16}$  on the tested range  $0.02 \leq t \leq 0.2$ .

The finite Dirichlet sum at  $s = 2.30 + 0.41i$  converges only algebraically; with 50,000 terms its discrepancy from  $\zeta(s+1)/(1-2^{-s})$  was  $1.15 \times 10^{-10}$ , consistent with the expected truncation tail. This check is deliberately performed without convergence acceleration.

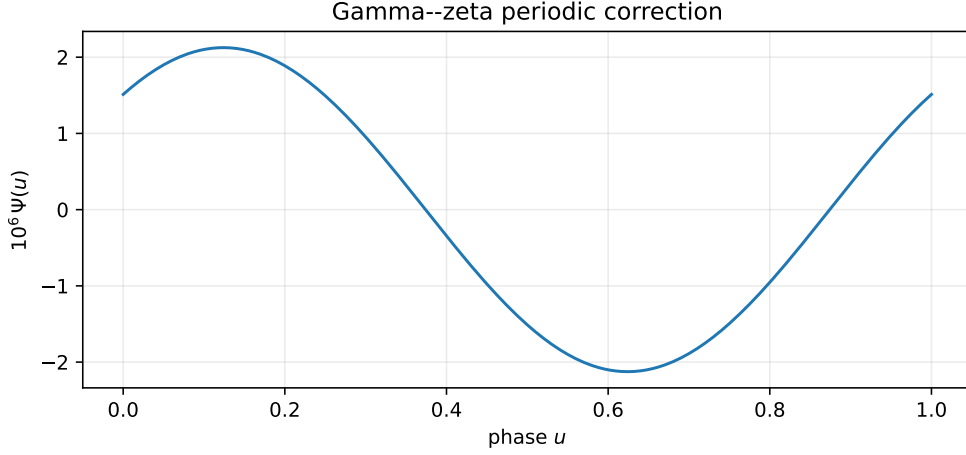


Figure 1: The centered periodic Gamma–zeta correction. Its amplitude is small, but it occurs at the same asymptotic order as the nonoscillatory constants and therefore cannot be dropped.

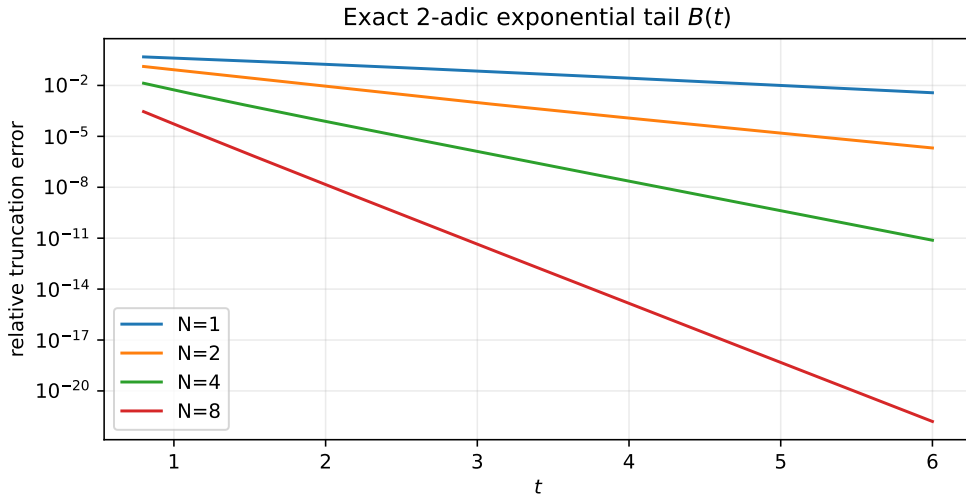


Figure 2: Relative error after retaining  $N = 1, 2, 4, 8$  terms of the exact 2-adic exponential series for  $B(t)$ . The bound (7.12) gives a simple rigorous enclosure.

## 10 Consequences and structural interpretation

### 10.1 There is no phase-free complete expansion

The first coefficient

$$g_1 = \frac{1}{2} \log \rho + \kappa_0 - \Psi(\rho + \beta)$$

already shows that a complete expansion cannot have constant coefficients in powers of  $\rho^{-1}$  and  $\log \rho$  alone. Formula (3.4) propagates phase derivatives to every later order. The correct coefficient ring is not

$\mathbb{R}[\log \rho]$ , but

$$\mathbb{R}[\log \rho] \otimes C^\infty(\mathbb{R}/\mathbb{Z}) \quad (10.1)$$

enlarged by the forward periodic saddle coefficients. The phase circle is an intrinsic asymptotic variable.

## 10.2 Why the highest logarithms are nevertheless elementary

Theorem 4.2 shows a clean separation:

$$\text{highest slow logarithm is universal, while phase information begins one degree lower.} \quad (10.2)$$

This reconciles two observations that can otherwise look contradictory. An elementary Lambert approximation captures the dominant logarithmic tower, but it cannot be promoted to a complete asymptotic expansion because the periodic coefficient at the next layer is nonconstant.

## 10.3 A common Gamma–zeta source

The denominator  $1 - 2^{-s}$  has poles at the complex dimensions  $2\pi i k/L$ . In the small- $t$  Mellin expansion of  $B$ , they produce  $-\Psi$ . In the exact inverse expansion, derivatives and products of  $\Psi$  are transported by Lagrange and Bell polynomials. Thus the entire hierarchy of inverse oscillations is generated from the same dyadic pole lattice; no new frequencies are introduced, only integer convolutions of the existing modes.

## 10.4 Arithmetic meaning of the ultraflat sectors

The integer weight

$$f(n) := nb_n = 2^{\nu_2(n)+1} - 1 \quad (10.3)$$

counts the geometric sum of all dyadic representations  $n = m2^k$ . It satisfies

$$f(2n) = 2f(n) + 1, \quad f(2n+1) = 1. \quad (10.4)$$

If  $C(z) = \sum_{n \geq 1} f(n)z^n$ , then

$$C(z) = 2C(z^2) + \frac{z}{1-z}. \quad (10.5)$$

This puts the exponentially small saddle correction in the same digital universe as the Thue–Morse and ruler sequences, but with a different arithmetic statistic: geometric accumulation over the full 2-adic valuation rather than parity of the binary digit sum.

# 11 Conjectures and research directions

The following statements are deliberately separated from the proved results. They are intended as concrete frontier problems rather than rhetorical speculation.

**Conjecture 11.1** (Uniform Gevrey bound for phase-aware inverse coefficients). *There exist constants  $C, A > 0$  and a phase-analytic strip  $|\Im \vartheta| < \sigma$  such that the weighted analytic norms of the inverse coefficients satisfy*

$$\|g_n\|_\sigma \leq CA^n n! \quad (11.1)$$

*when the polynomial dependence on  $\ell$  is measured with the natural weight  $1 + |\ell|$ . The same Gevrey order should hold for  $d_n$  and  $h_n$ .*

A proof would require corresponding large-order control of the forward  $A_j$ , followed by norm estimates for the closed Lagrange extractor. The finite formula (3.4) is well suited to such estimates.

**Conjecture 11.2** (Sharp Fourier strip inheritance). *For every fixed  $n$ , each periodic coefficient of  $d_n, h_n, g_n$  extends to the same maximal horizontal strip as  $\Psi$ , and no generic cancellation enlarges that strip. In particular, the exponential rate of Fourier decay is stable under all inverse coefficient operations.*

Differentiation, multiplication, and Bell composition preserve any smaller strip. The nontrivial part is sharpness: one must locate the nearest complex singularities and exclude cancellation among the many differential-polynomial terms.

**Conjecture 11.3** (Two-level exponentially improved inverse transseries). *For  $0 < y \ll 1$ , let  $q_*(y)$  be the exact positive saddle coordinate defined implicitly by*

$$G(y) = x(q_*(y)), \quad (11.2)$$

where  $x(q)$  is the exact saddle map (7.21). After optimal truncation of the algebraic phase-aware expansion,  $G(y)$  admits an exponentially improved description with at least two distinct levels: ordinary saddle-resurgent sectors controlled by the large-order growth of the  $A_j$ , and ultraflat dyadic sectors generated by

$$\exp(-n2^{q_*(y)}), \quad n \geq 1, \quad (11.3)$$

whose arithmetic seed is  $b_n = (2^{\nu_2(n)+1} - 1)/n$ .

The algebraic saddle inversion gives  $q_*(y) = \vartheta + \mathcal{O}((\log \rho)/\rho)$ , but this estimate does *not* permit replacing  $q_*$  by  $\vartheta$  in (11.3): the induced change in the exponent is of order  $2^\vartheta(\log \rho)/\rho$ , which diverges. Thus the exact saddle coordinate is part of the beyond-all-orders datum, not merely a cosmetic refinement. Likewise, a naive substitution of (7.10) into the ordinary saddle series is not uniform because each derivative introduces powers of  $2^{q_*}$ ; see (7.23). A correct theory likely needs an accelerated or multilevel transseries in which  $\rho^{-1}$  and  $\exp(-2^{q_*(y)})$  are separate asymptotic monomials.

**Conjecture 11.4** (Resurgent arithmetic of the dyadic sectors). *Within the ultraflat level, the Stokes or connection coefficients are generated by Dirichlet convolutions of the sequence  $b_n$ , so their Dirichlet series are finite products of  $\zeta(s+1)/(1-2^{-s})$  and shifted variants.*

This is motivated by the fact that nonlinear saddle and inverse operations multiply exponential sectors, turning coefficient sequences into additive convolutions. The conjecture is precise enough to test computationally once a uniform hyperasymptotic construction is available.

## 11.1 Promising concrete projects

1. **Generate  $A_2, A_3, A_4$  in compact Fourier-differential form.** The partition formula (6.5) removes one combinatorial bottleneck. Combining it with the exact  $q$ -to- $\lambda$  conversion would make  $g_3, g_4, g_5$  completely explicit in  $\Psi$  alone.
2. **Prove weighted Wiener-algebra bounds.** Because the Fourier coefficients of  $\Psi$  decay exponentially, a weighted  $\ell^1$  Fourier algebra should control products, derivatives, Lagrange composition, and Bell polynomials uniformly in phase.
3. **Develop optimal truncation experiments.** High-precision values of  $G(y)$  can be compared with successive algebraic truncations at fixed phase sequences. The error should reveal whether the first visible beyond-all-orders scale is saddle-resurgent or the much smaller dyadic scale  $\exp(-2^{q_*(y)})$ .
4. **Base- $b$  generalization.** For a geometric scale  $b^{-j}$ , the analogue of (7.10) is expected to have coefficients

$$\frac{b^{\nu_b(n)+1} - 1}{(b-1)n},$$

and the same Dirichlet kernel  $\zeta(s+1)/(1-b^{-s})$ . The resulting inverse phase would have period one in  $\log_b$ .

5. **Lean formalization.** The most modular targets are the generalized Lagrange formula (3.4), the Bell extraction (4.6), the regrouping proof of (7.10), and the Dirichlet identity (7.16). Each can be formalized independently of delicate contour estimates.

## 12 Formalization blueprint

A Lean development can be divided into four layers.

**Formalization obligation 12.1** (Formal two-scale coefficient ring). Define truncated series in  $z$  with coefficients in a differential algebra of polynomials in  $\ell$  and smooth periodic functions of  $\vartheta$ . Prove that Taylor composition, inversion of  $1 + O(z)$ , logarithm, and exponential preserve truncation equivalence.

**Formalization obligation 12.2** (Parameterized Lagrange inversion). For a commutative  $\mathbb{Q}$ -algebra and  $W = t\Phi(z, W)$ , prove

$$[t^k]W = \frac{1}{k}[u^{k-1}]\Phi(z, u)^k.$$

Specialize  $t = z$  and prove (3.4). This isolates the only nontrivial formal-series theorem needed for the closed inverse coefficients.

**Formalization obligation 12.3** (Gaussian contraction). Represent a Gaussian moment functional algebraically by  $\mathbb{E}Z^{2m} = (2m - 1)!!$  and  $\mathbb{E}Z^{2m+1} = 0$ . Expand (6.3) in a truncated graded algebra and prove the constrained sum (6.5).

**Formalization obligation 12.4** (2-adic regrouping). Formalize the bijection

$$\{(k, m) : k \geq 0, m \geq 1, m2^k = n\} \longleftrightarrow \{0, 1, \dots, \nu_2(n)\}$$

and sum the resulting geometric progression. Absolute convergence then justifies rearrangement in (7.10); a second Tonelli argument proves (7.16).

The contour-shift portion of [theorem 7.5](#) is analytically deeper, but it is not needed to formalize the exact arithmetic identities or the closed inverse coefficient algebra.

## 13 Conclusion

The small-argument inverse Fabius function has a complete asymptotic expansion, but its natural coefficient space is intrinsically two-scale. The slow variable is  $\rho^{-1} \asymp (\log(1/y))^{-1/2}$ , the slow logarithm is  $\log \rho$ , and the fast variable is the exact phase  $\rho + 1/\log 2 + 1/2$  modulo one. The repository had already established this architecture and a recursive all-orders inversion. The present report makes the arbitrary-order coefficient genuinely explicit:

- generalized Lagrange–Bürmann gives  $d_n$  directly;
- partial Bell polynomials give  $g_n$  directly;
- the leading logarithm is universally  $(\log \rho)^n / (2^n n!)$ ;
- Gaussian contractions admit a finite constrained-partition formula;
- inverse derivatives follow from one two-scale operator.

The exact dyadic Bose tail adds a second structural layer. Its coefficients are controlled by  $\nu_2(n)$ , its Mellin transform contains the same Gamma–zeta complex dimensions as the periodic phase, and its derivative is the exact exponentially small correction to the saddle map. This turns a formal “beyond all orders” remainder into an explicit arithmetic object and provides a concrete route toward an exponentially improved inverse theory.

## A A compact coefficient dictionary

For convenient implementation, the full direct construction can be summarized as follows:

$$\begin{aligned}
L &= \log 2, & \beta &= L^{-1} + \frac{1}{2}, & T &= \log(1/y), \\
\rho &= \sqrt{2T/L}, & z &= \rho^{-1}, & \ell &= \log \rho, \\
\vartheta &= \rho + \beta, & G_0 &= \rho e^{-L\rho - L\beta}, & \lambda &= \rho + \beta + D, \\
D &= \sum_{n \geq 1} d_n z^n, & d_n &= \sum_{k=1}^n \frac{(-1)^k}{kL^k} [z^{n-k} u^{k-1}] \mathcal{R}^k, \\
\frac{G}{G_0} &= (1 + \beta z + zD) e^{-LD}, & g_n &= [z^n] (1 + \beta z + zD) e^{-LD}.
\end{aligned}$$

The residual  $\mathcal{R}$  is given by (3.1). These lines are sufficient to implement the complete inverse expansion once the forward periodic coefficients  $A_j$  are available.

## B First generated saddle coefficients

For reference, the program generated the following logarithmic second saddle coefficient:

$$\begin{aligned}
\Lambda_2 &= \frac{5}{16} \rho_3^4 + \frac{5}{8} \rho_3^3 - \frac{25}{48} \rho_3^2 \rho_4 + \frac{9}{8} \rho_3^2 - \frac{2}{3} \rho_3 \rho_4 + \frac{7}{48} \rho_3 \rho_5 + 2 \rho_3 \\
&\quad + \frac{1}{12} \rho_4^2 - \frac{1}{2} \rho_4 + \frac{1}{8} \rho_5 - \frac{1}{48} \rho_6 + \frac{5}{2}.
\end{aligned} \tag{B.1}$$

Higher generated expressions are archived separately rather than typeset in full because their term count grows rapidly.

## C Reproduction instructions

From the extracted archive directory, run

```
python inverse_fabius_experiments.py --max-saddle-order 3
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  inverse_fabius_complete_asymptotics.tex
```

The Python program regenerates `generated_coefficients.tex`, `verification_output.txt`, and the two vector PDF figures. Increasing `--max-saddle-order` generates further  $C_n$  and  $\Lambda_n$ ; symbolic size, rather than numerical instability, is the main limitation.

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